Asthma and Respiratory Immunology

Companion Figures

24 figures, ordered as referenced in the .md.

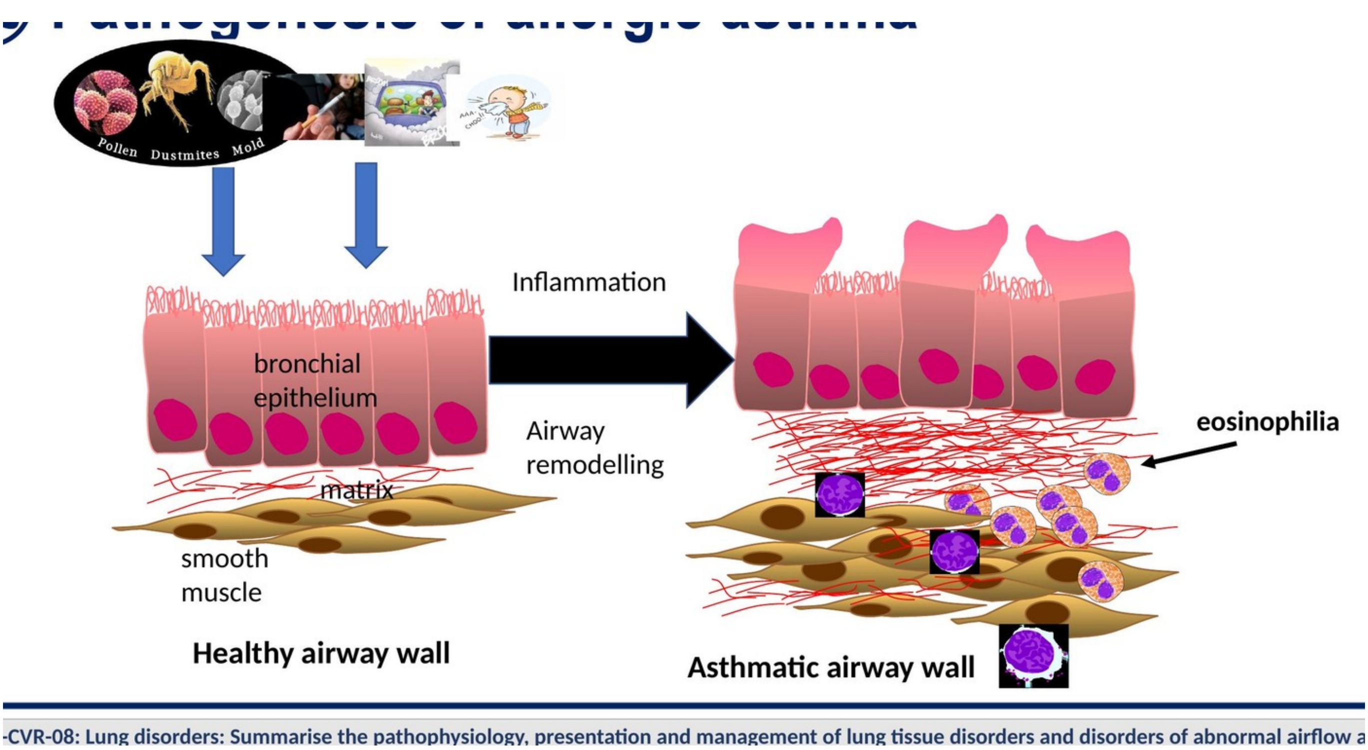


Figure L1.1 — Healthy vs asthmatic airway wall: epithelium, matrix, smooth muscle, eosinophils

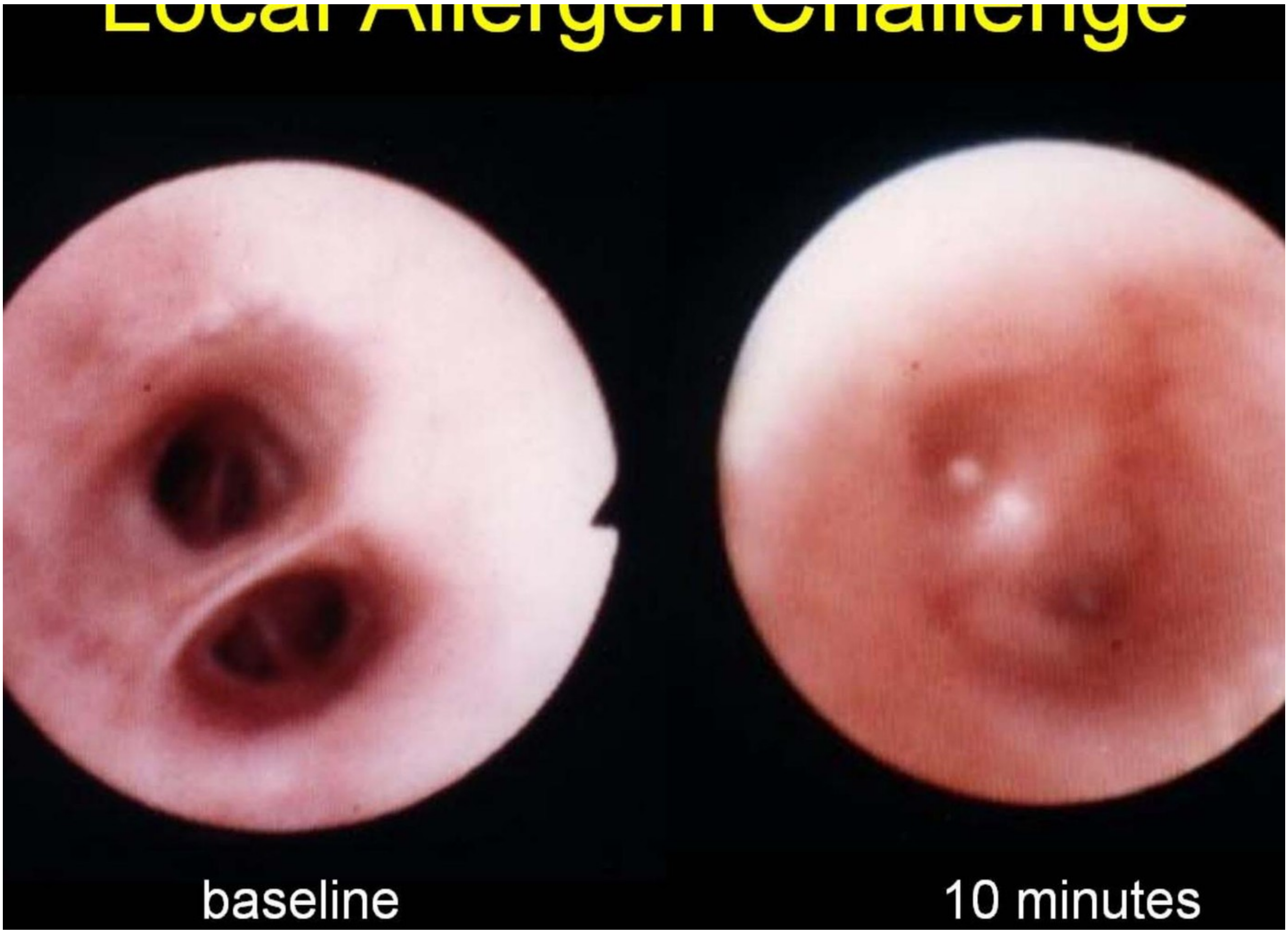


Figure L1.2 — Local allergen challenge at 10 minutes (early-phase bronchoconstriction)

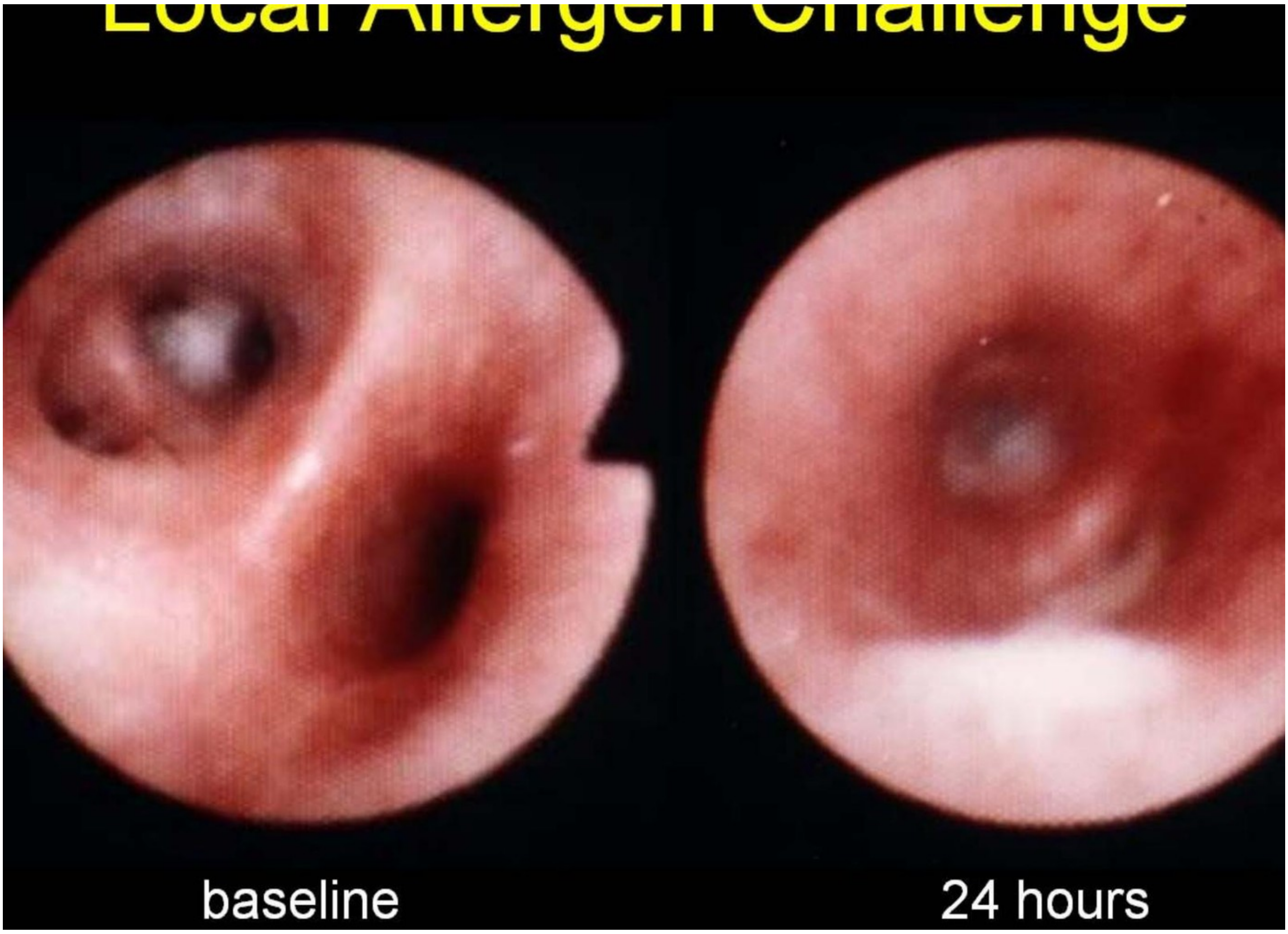


Figure L1.3 — Local allergen challenge at 24 hours (late-phase persistent narrowing)

Reversible airflow obstruction

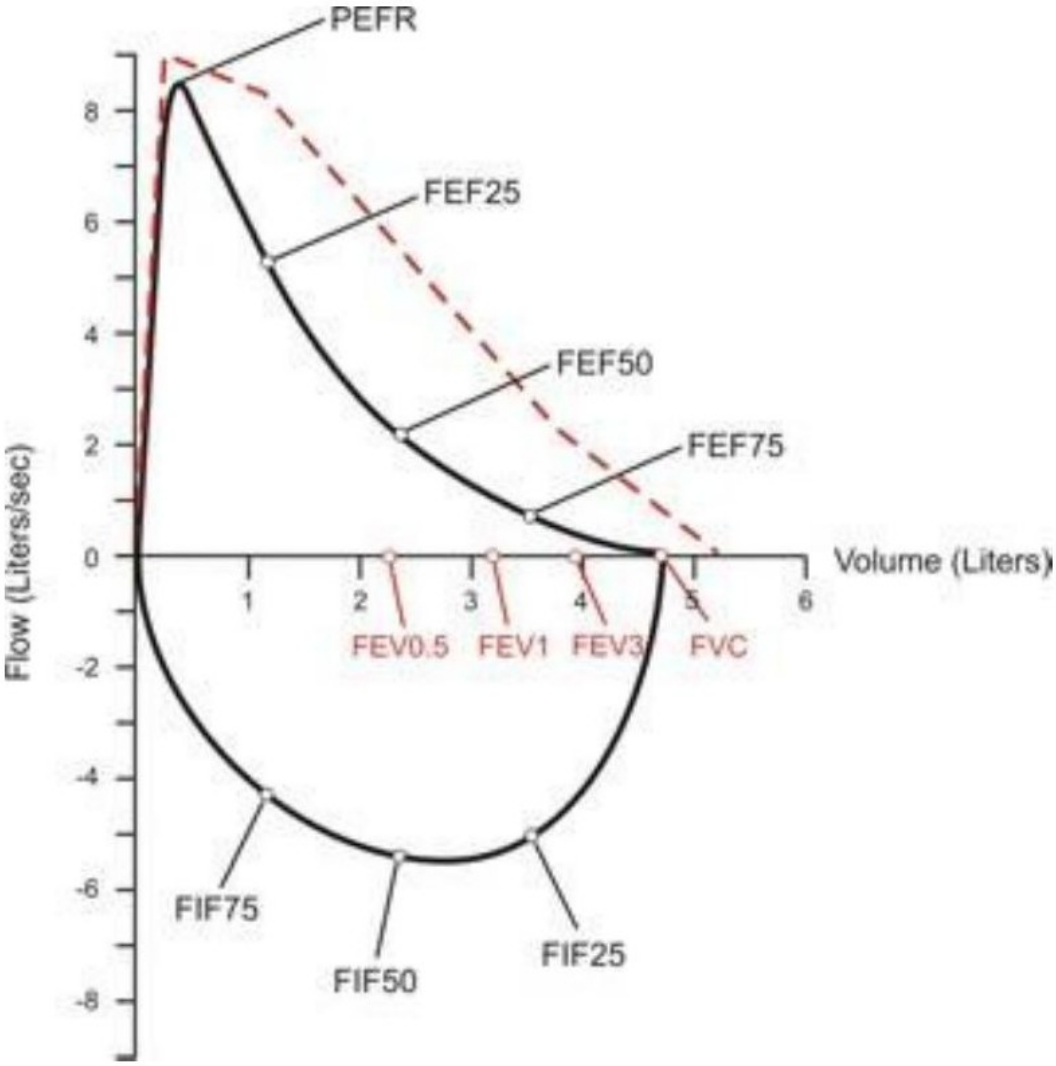


Figure L1.4 — Asthmatic flow-volume loop: scooped expiratory curve

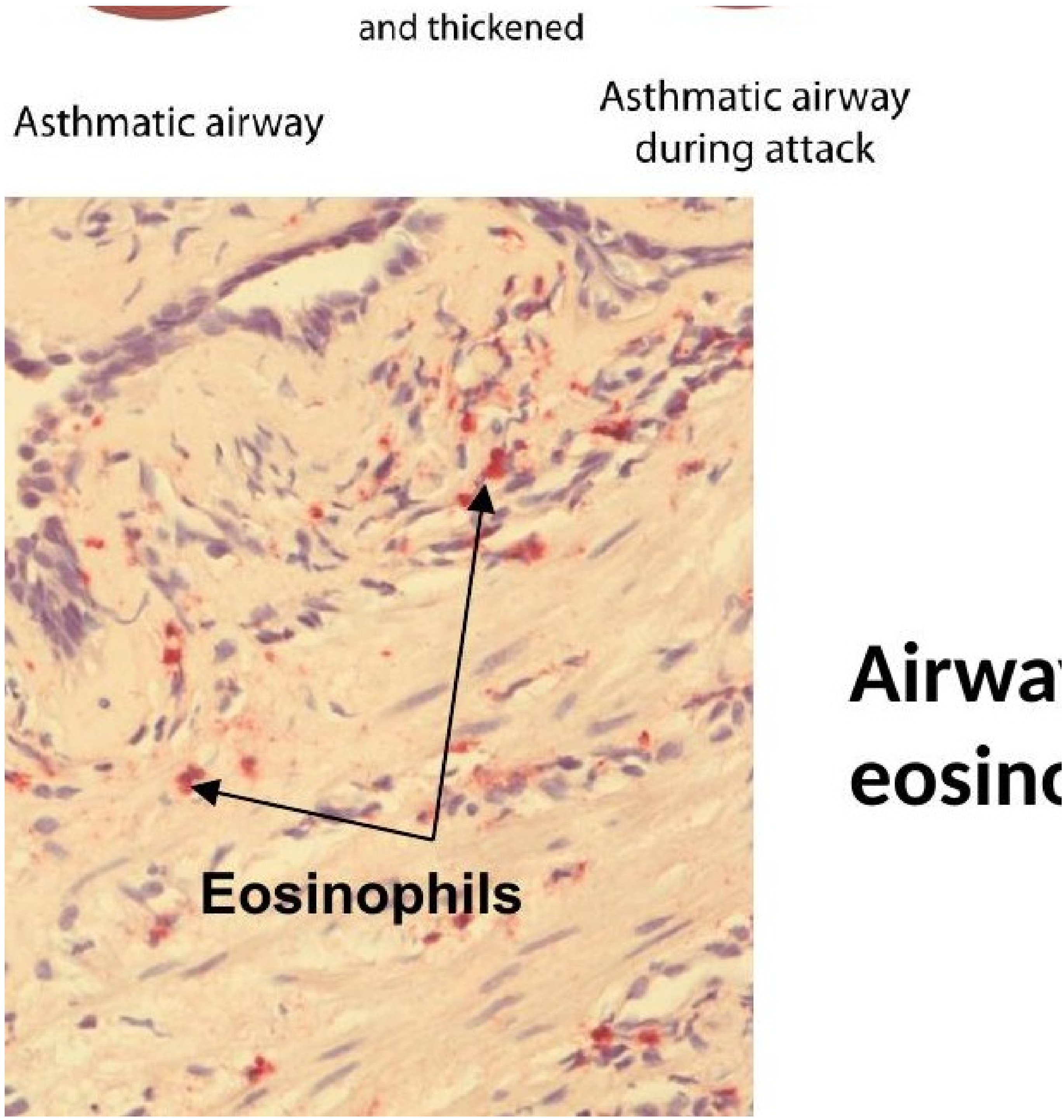


Figure L1.5 — Airway histology: eosinophil infiltrate

Eosinophilia

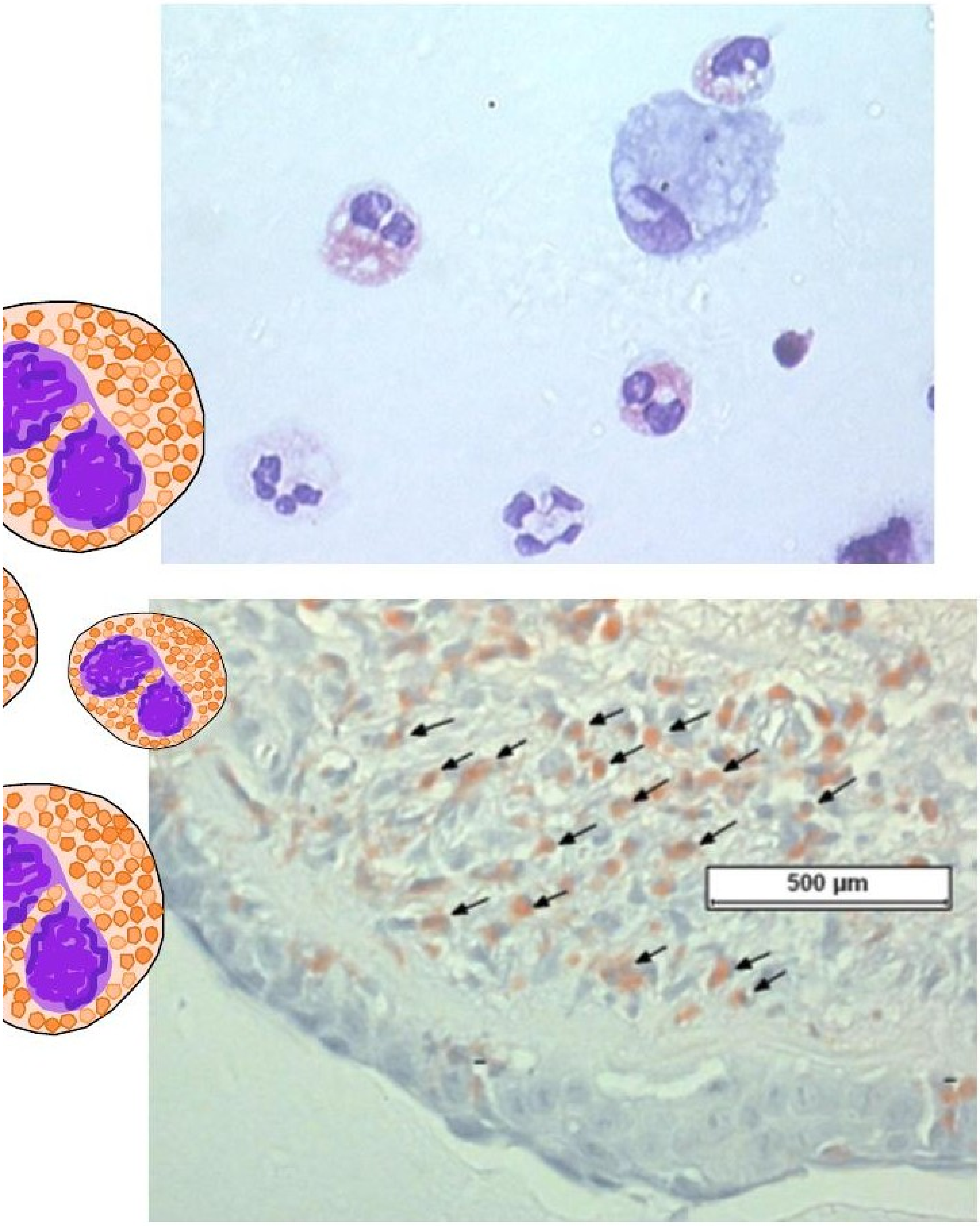
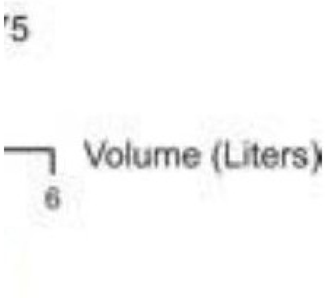
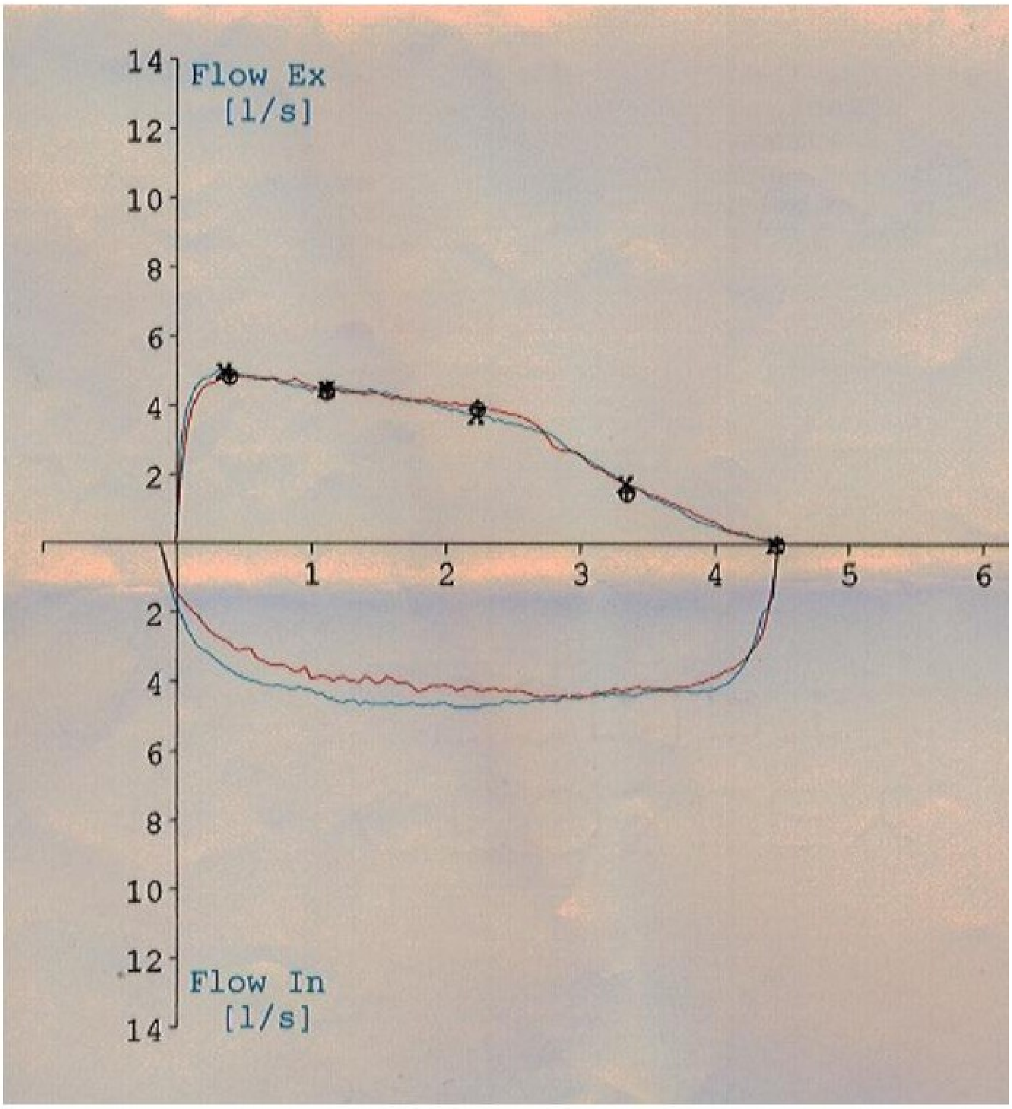


Figure L1.6 — Eosinophil morphology (bilobed nucleus, granules) + airway tissue infiltrate

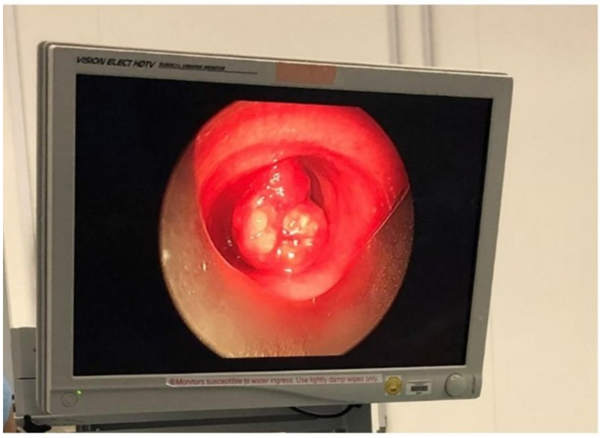




Upper/large airway fixed obstruction – NOT asthma

Figure L1.7 — Flat-topped flow-volume loop = fixed upper-airway obstruction (NOT asthma)

Endotracheal tumour



Large airway foreign body

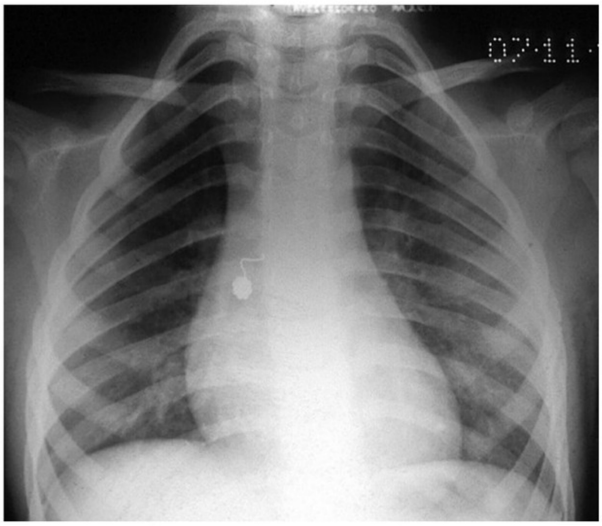


Figure L1.8 — Differentials for wheeze: endotracheal tumour and inhaled foreign body

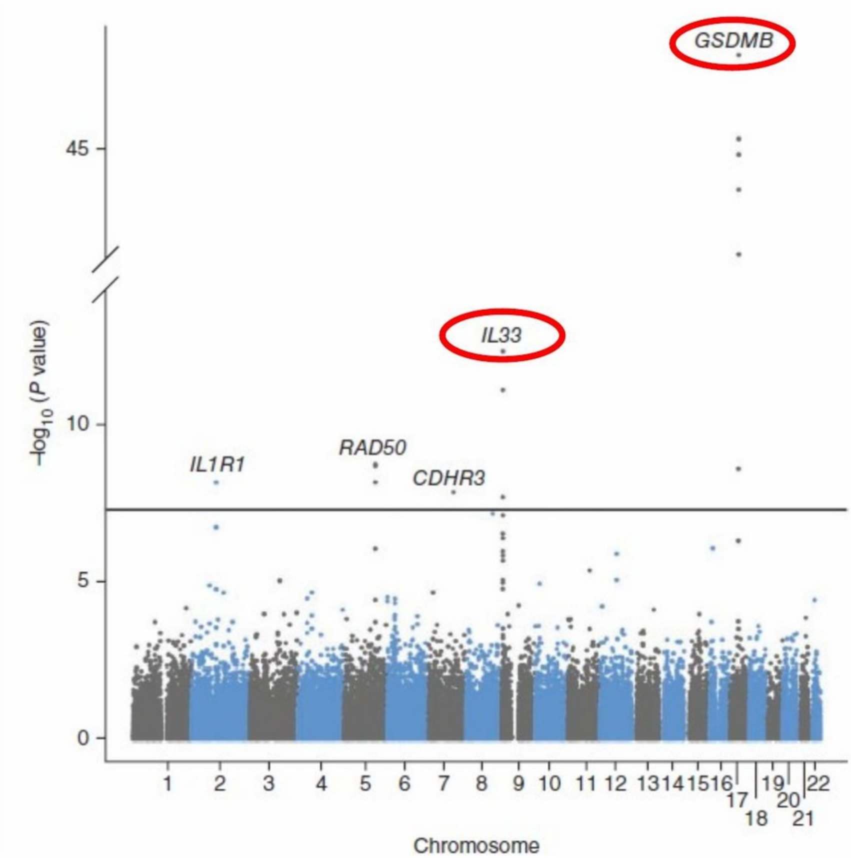


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Figure L1.9 — GWAS Manhattan plot for asthma susceptibility (GSDMB, IL33, CDHR3)

...mation in allergic asthma

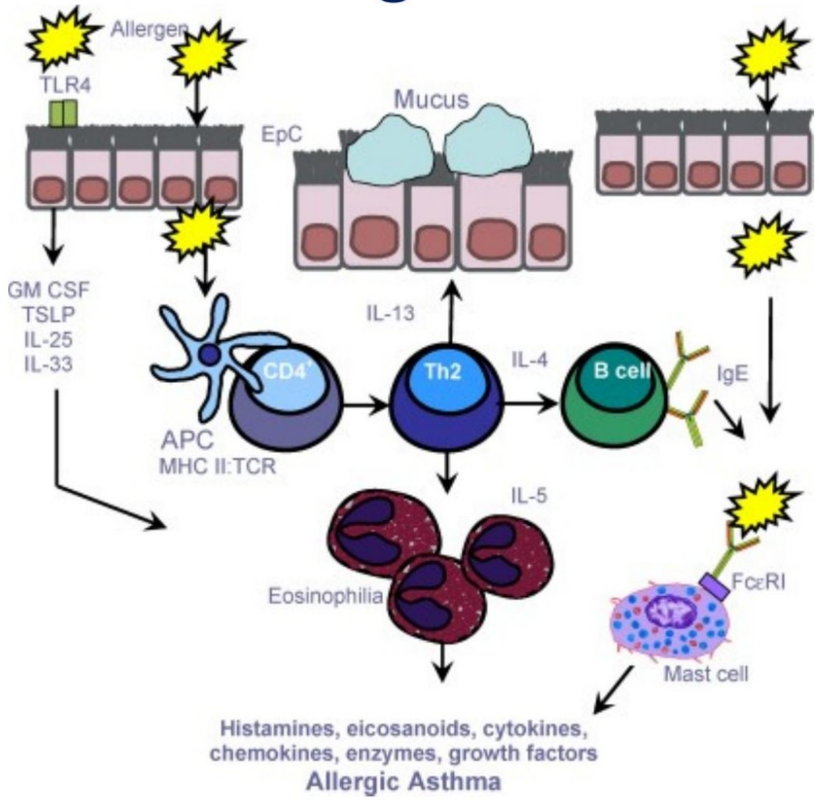

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Figure L1.10 — Allergic asthma immunopathology: epithelial alarmins, APC/Th2 axis, eosinophil, IgE/mast cell

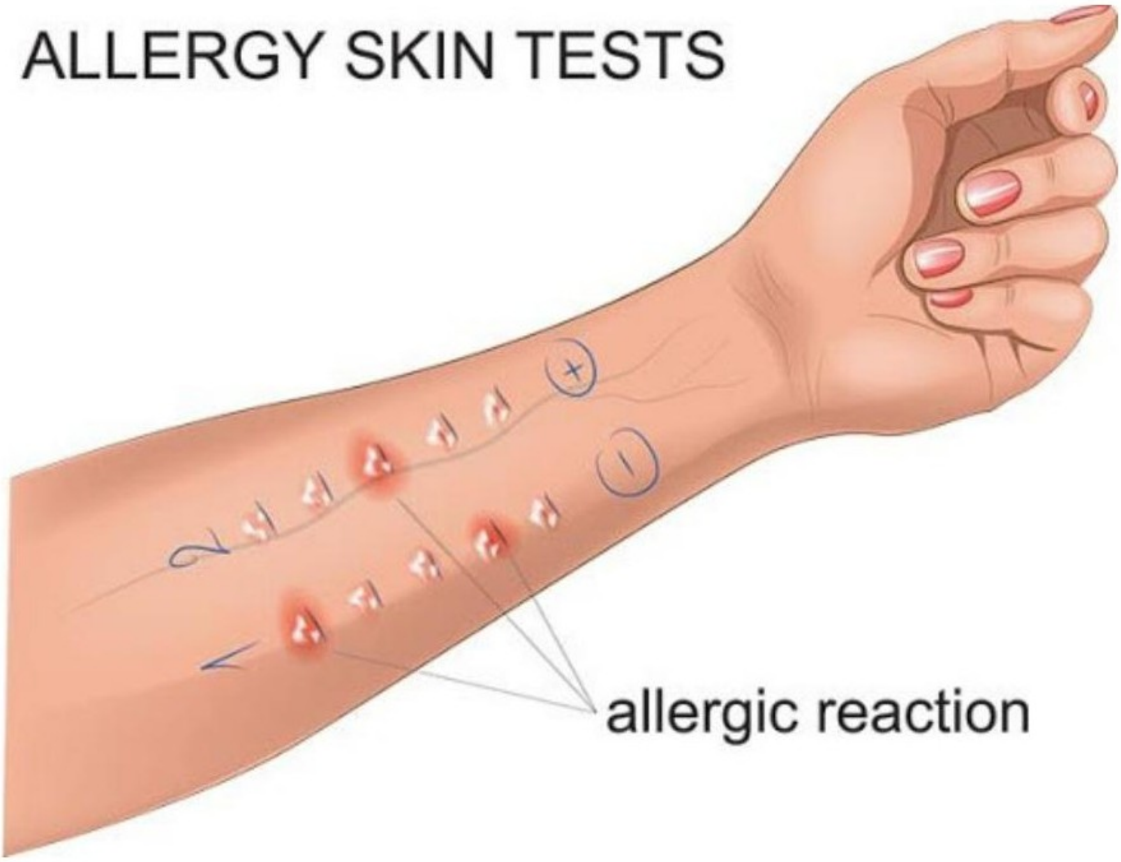


Figure L1.11 — Skin prick test: positive vs negative reactions

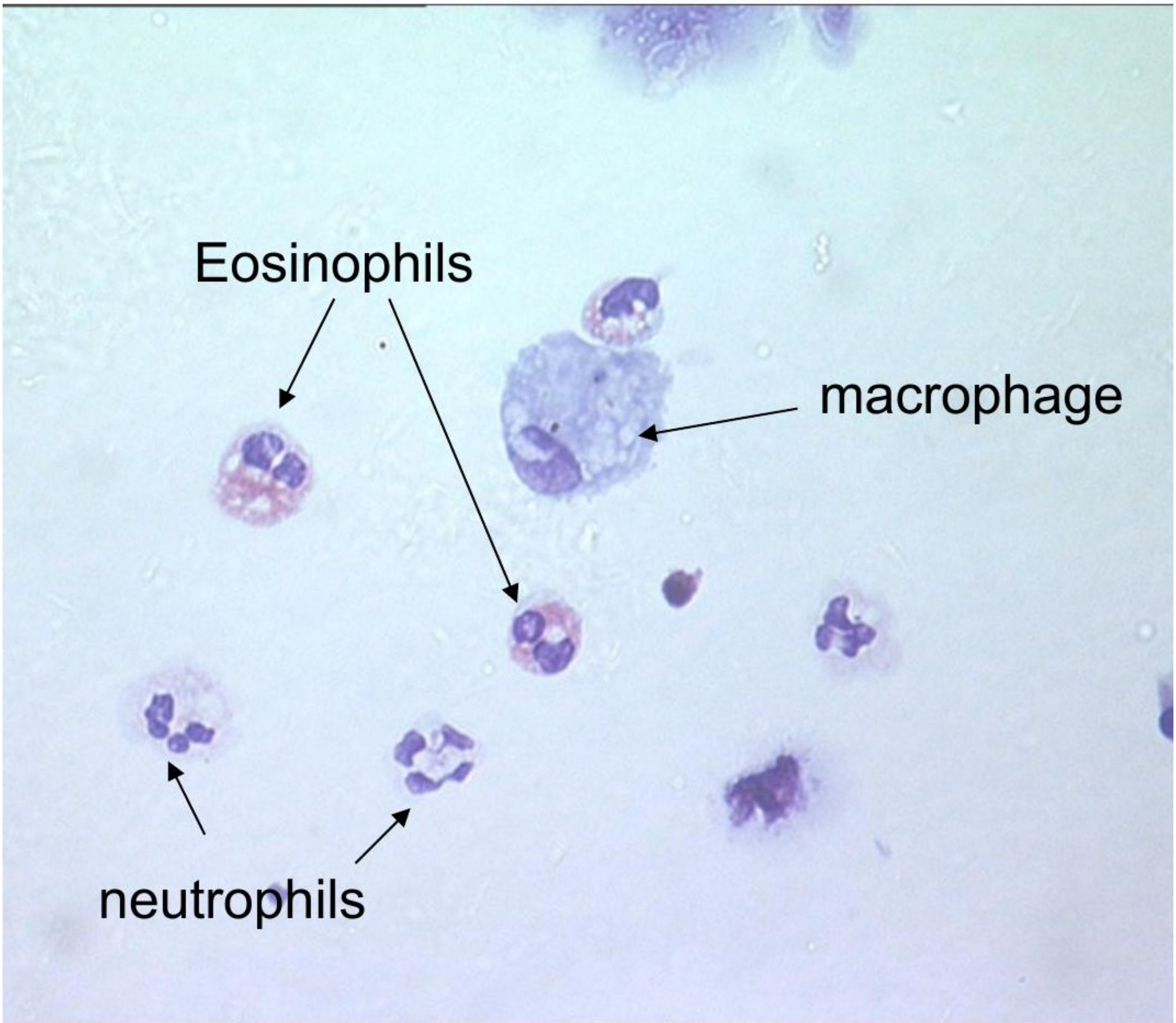


Figure L1.12 — Sputum cytology: eosinophils, neutrophils, macrophage

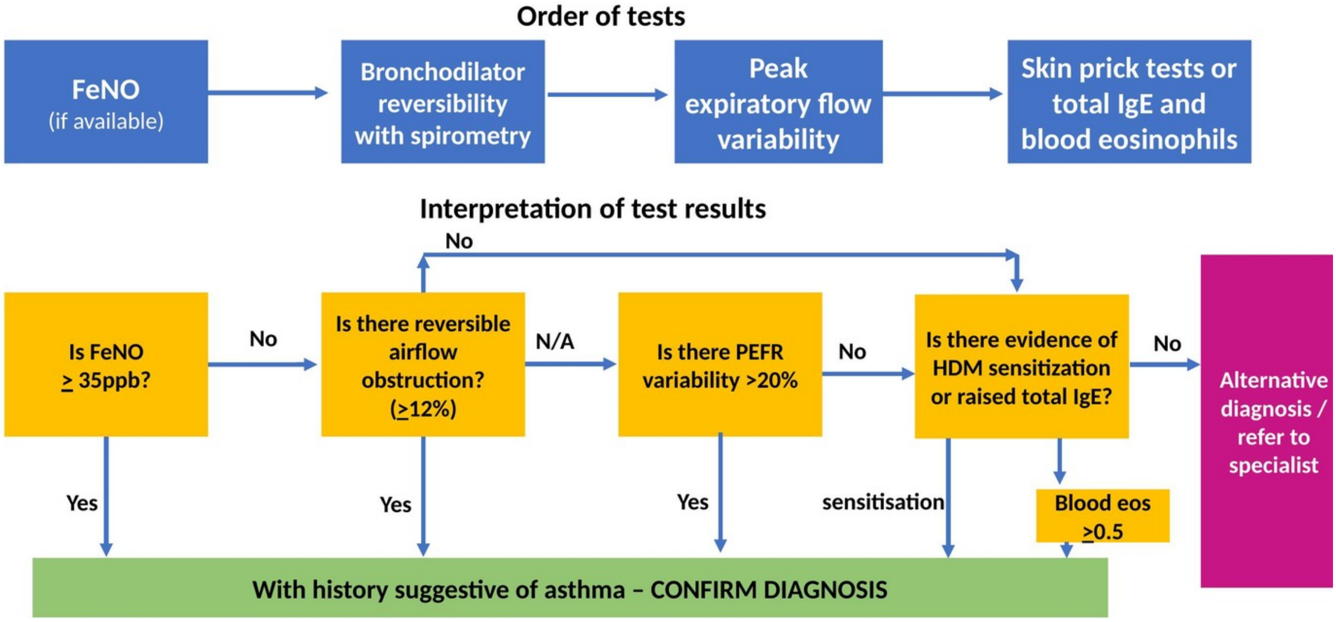


Figure L1.13 — NICE 2024 asthma diagnosis algorithm: children 5–16 years

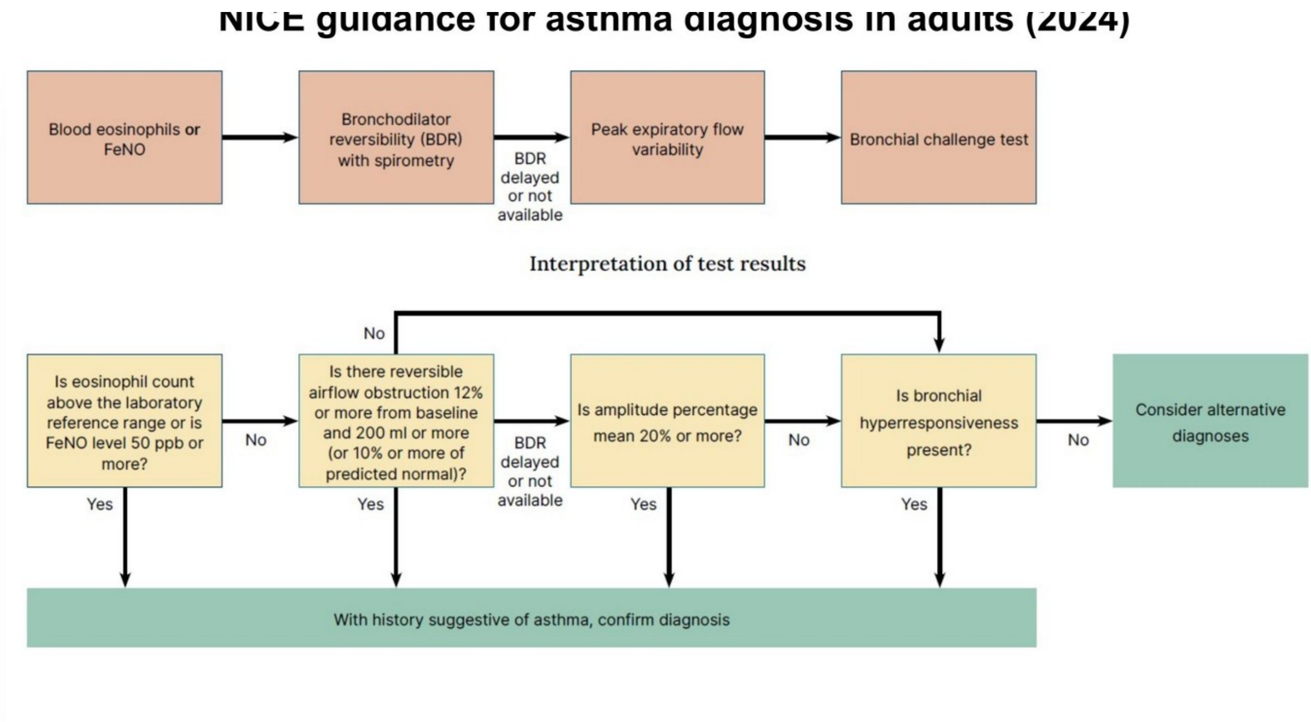


Figure L1.14 — NICE 2024 asthma diagnosis algorithm: adults

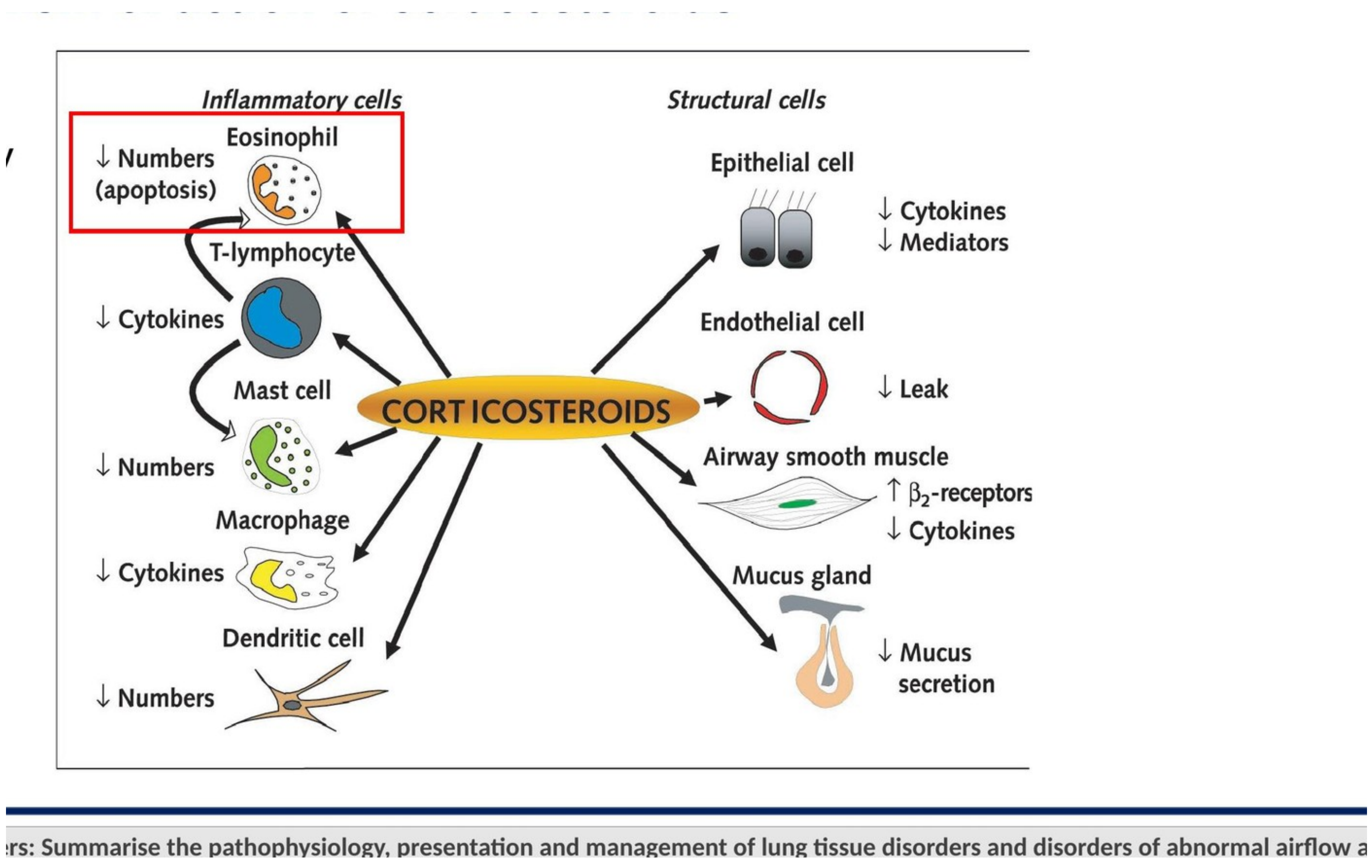


Figure L1.15 — Mechanism of corticosteroids on inflammatory and structural airway cells

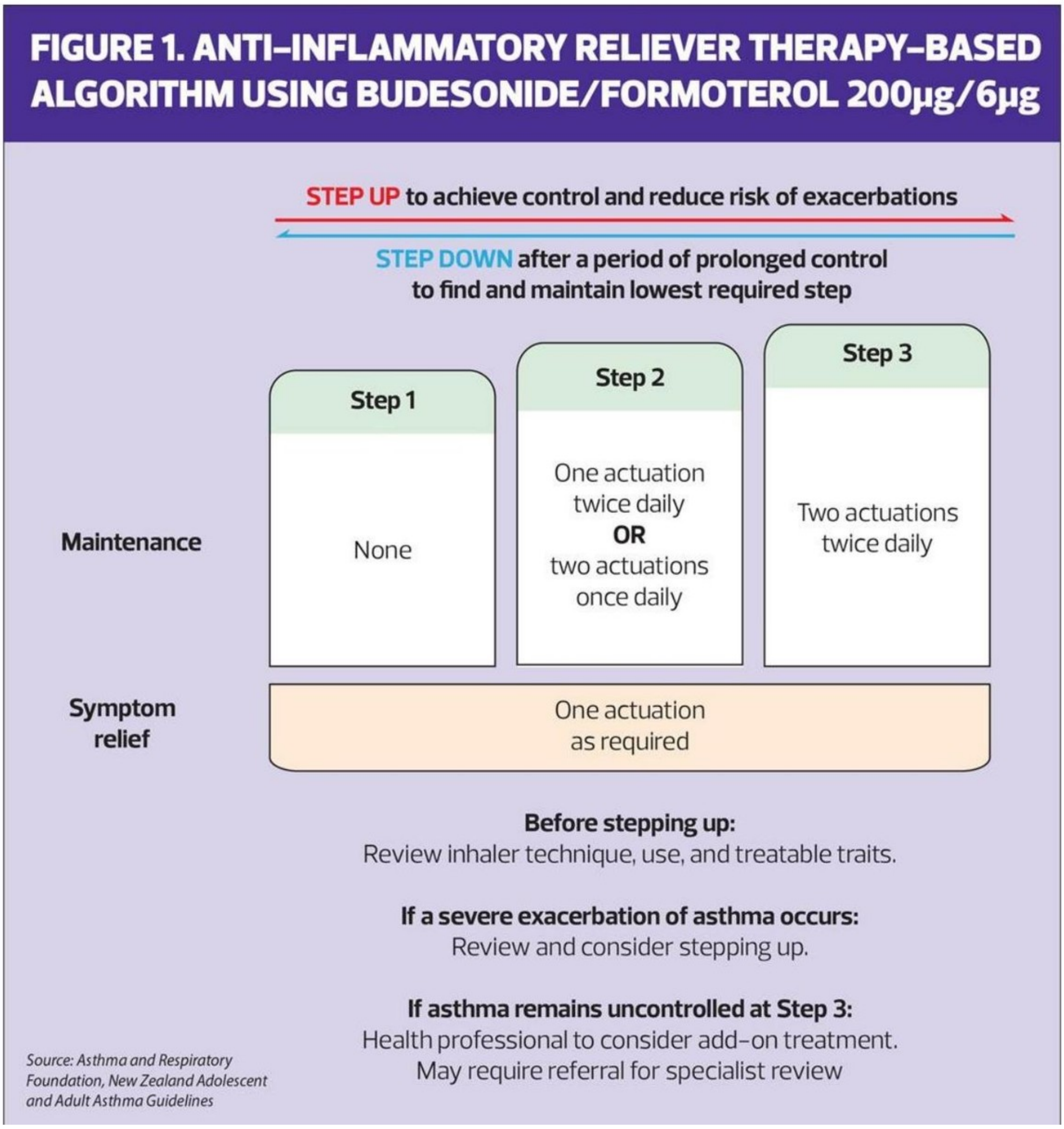


Figure L1.16 — Anti-inflammatory reliever (AIR) therapy stepwise algorithm: budesonide/formoterol 200/6 μg

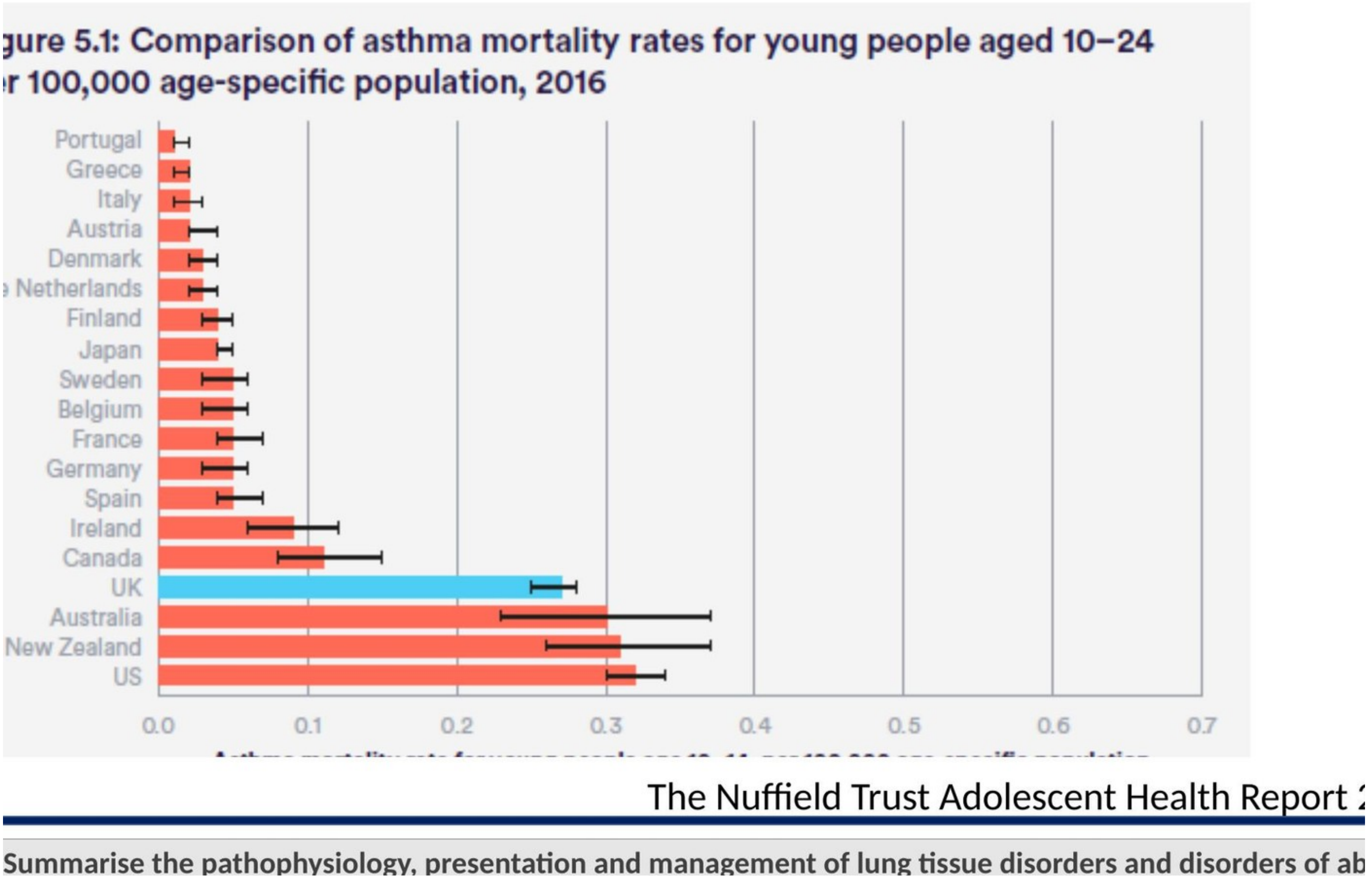


Figure L1.17 — Asthma mortality 10–24 years (Nuffield Trust 2019): UK 3rd-highest

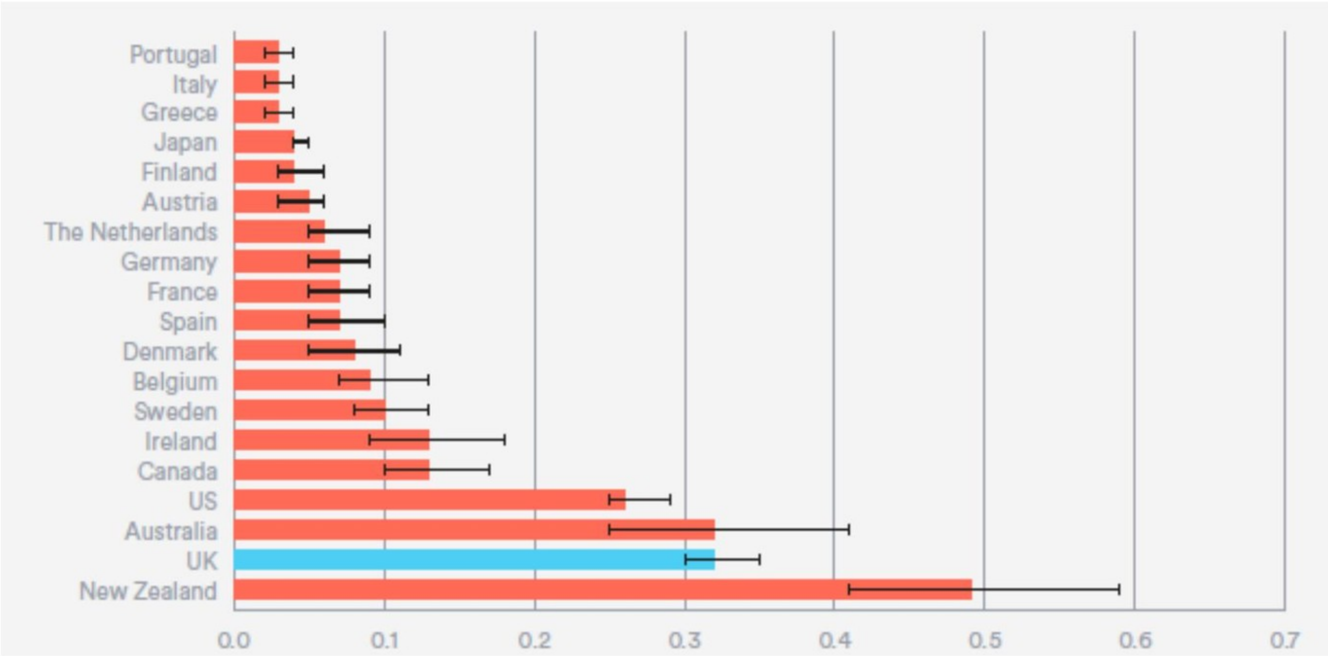


Figure L1.18 — Asthma mortality 10–14 years: UK 2nd-highest

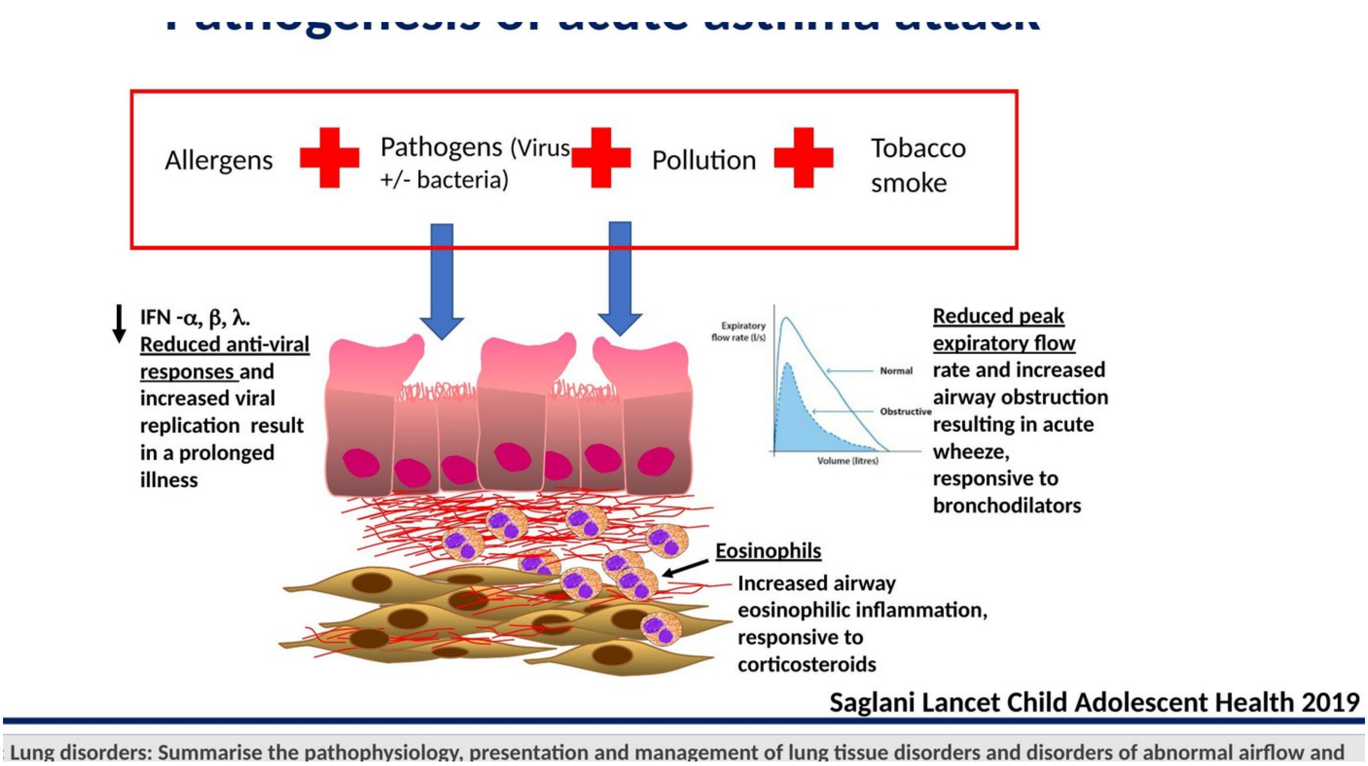
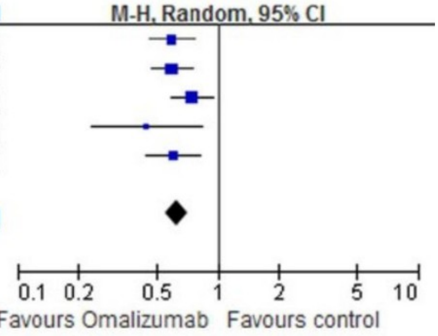


Figure L1.19 — Pathogenesis of acute asthma attack: triggers, reduced IFN, eosinophilia, obstruction



tudy or Subgroup	Favours Omalizumab Events	Total	Control Events	Total	Weight	Risk Ratio M-H, Random, 95% CI
uhl 2002 [47]	61	254	93	229	22.6%	0.59 [0.45, 0.77]
usse 2011 [49]	63	208	108	211	27.3%	0.59 [0.46, 0.76]
anier 2003 [45]	78	245	92	215	28.5%	0.74 [0.59, 0.95]
anier 2009 [48]	17	421	19	207	4.1%	0.44 [0.23, 0.83]
liven 2008 [46]	45	115	32	49	17.5%	0.60 [0.44, 0.81]
otal (95% CI)		1243		911	100.0%	0.63 [0.55, 0.71]
otal events	264		344			

leterogeneity: Tau² = 0.00; Chi² = 3.65, df = 4 (P = 0.46); I² = 0%
est for overall effect: Z = 7.19 (P < 0.00001)

2 | The effect of omlizumab on asthma exacerbations.

- **Only approx. 60% of patients are eligible**
- **Of those, only approx. 50-60% respond**

Figure L1.20 — Omalizumab vs placebo: pooled exacerbation risk ratio 0.63 (Lai 2015)

Mepolizumab: reduction in a

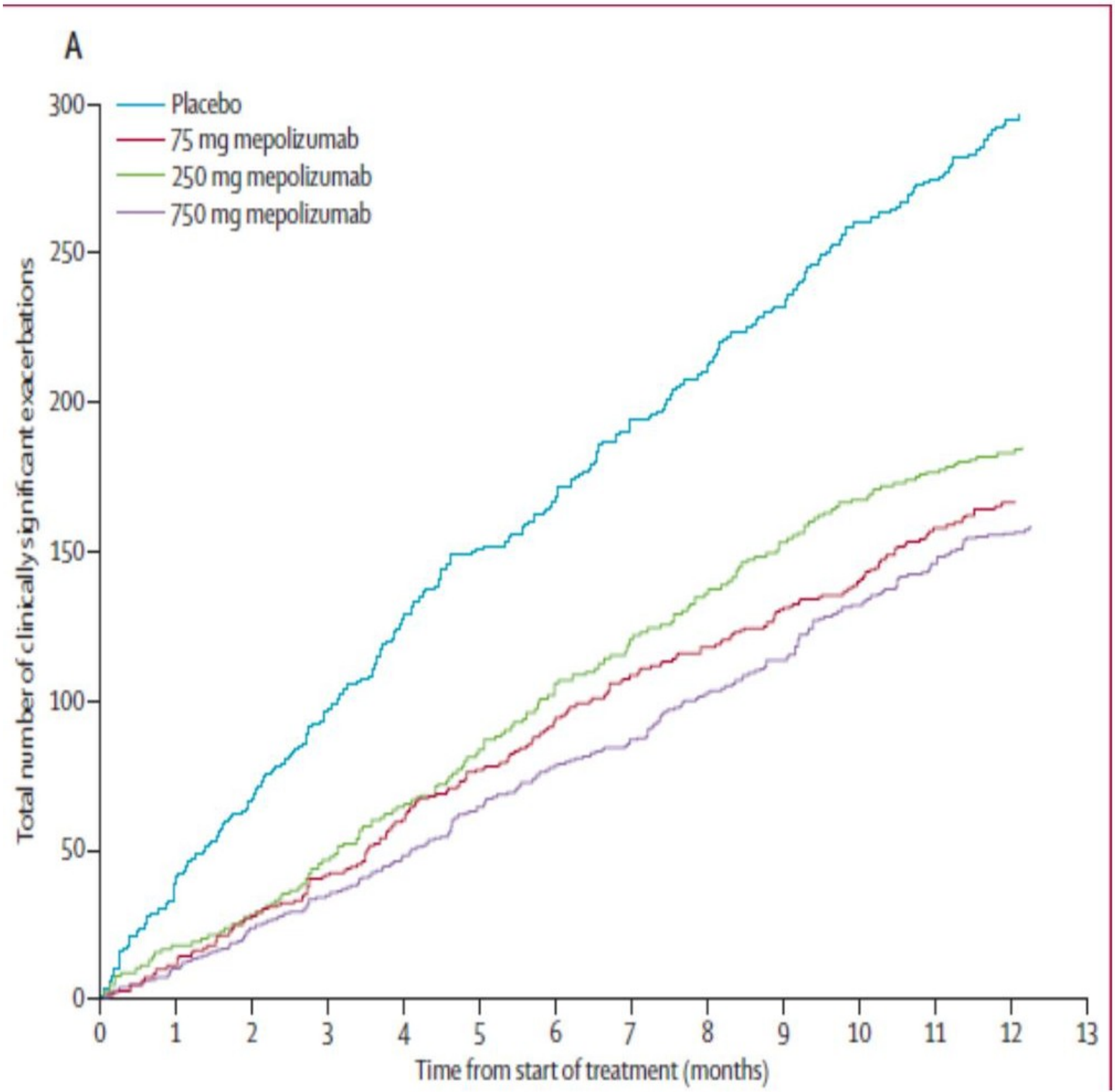


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Figure L1.21 — Mepolizumab vs placebo: cumulative exacerbations in adults (Pavord 2012)

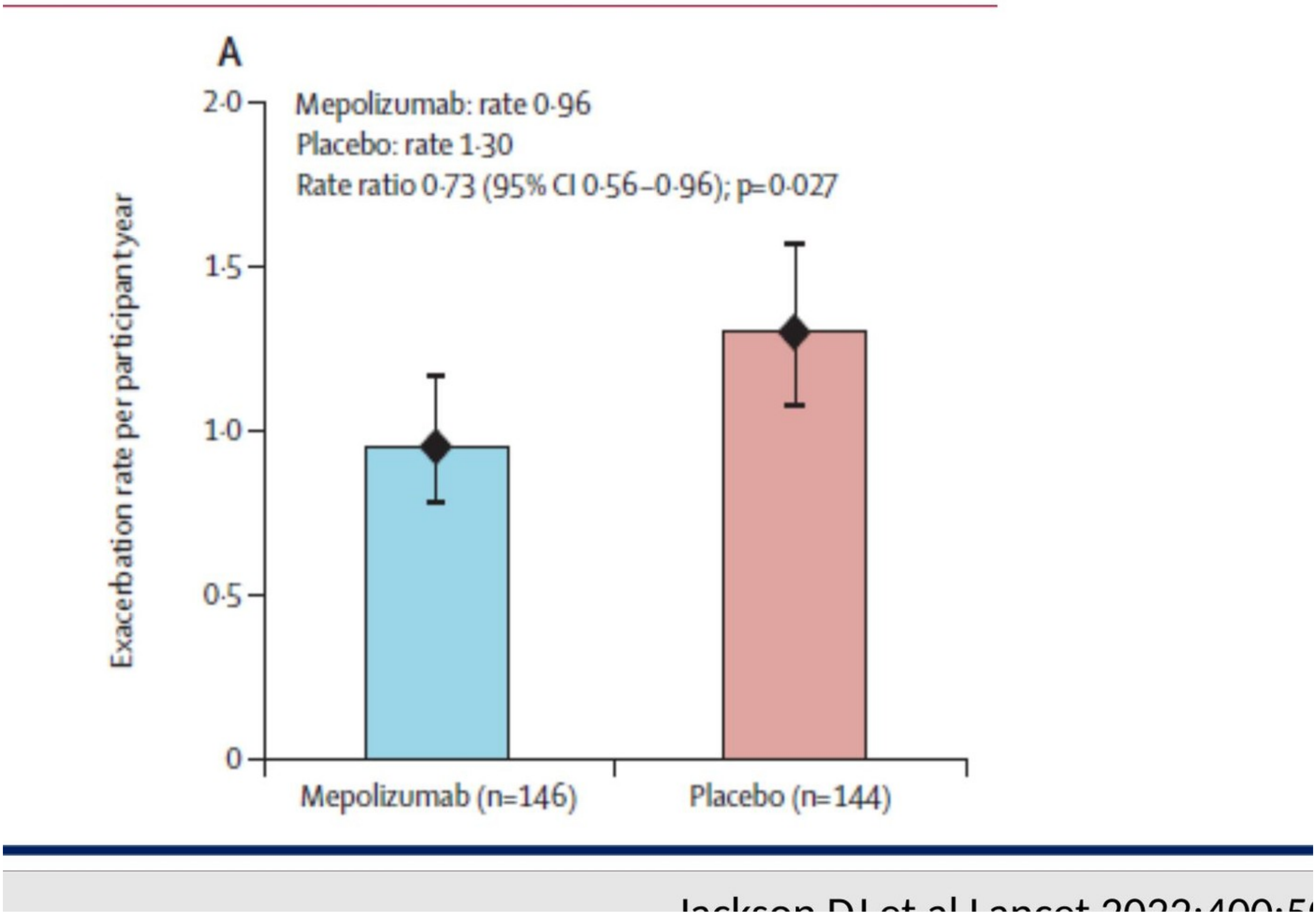

Figure L1.22 — Mepolizumab vs placebo: paediatric exacerbation rate (Jackson 2022)

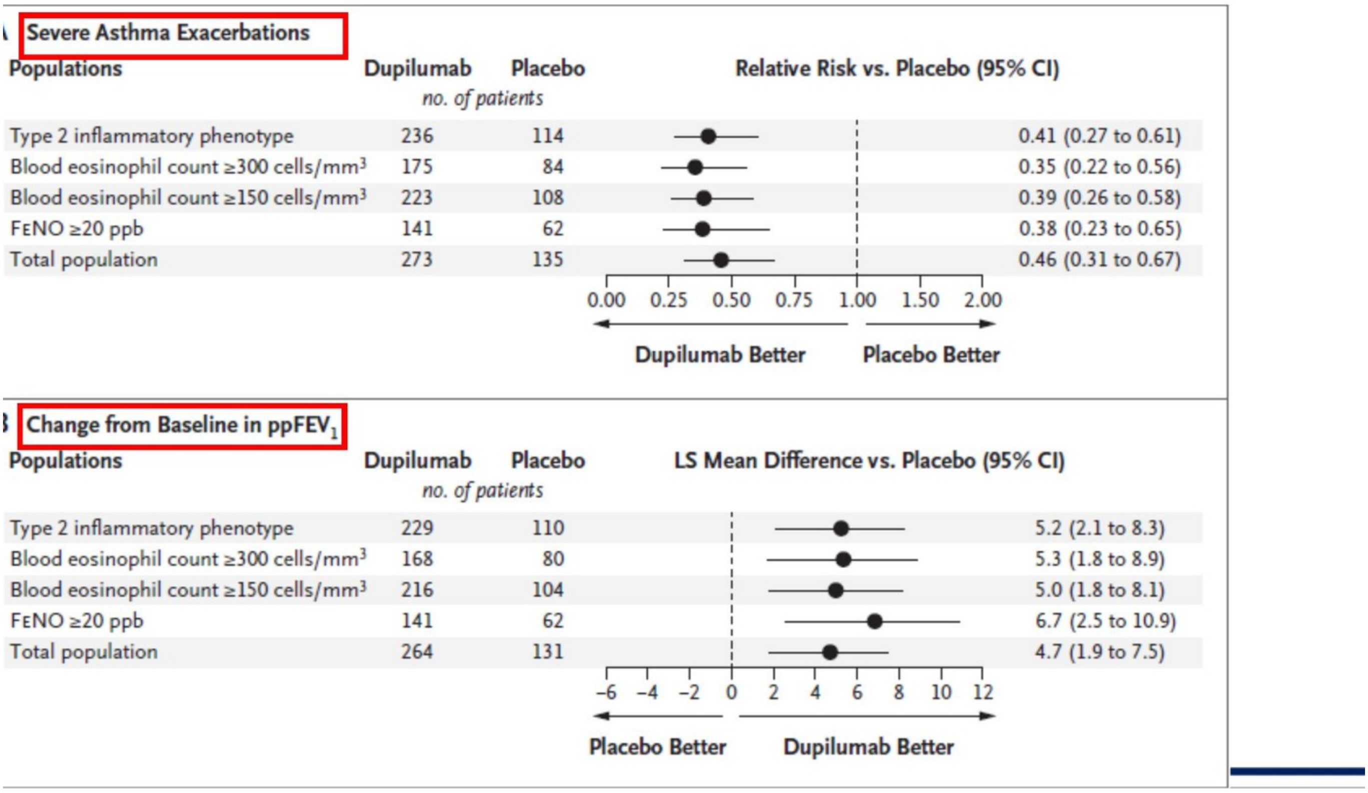

Figure L1.23 — Dupilumab vs placebo: exacerbations and pp-FEV1 improvement (Bacharier 2021)

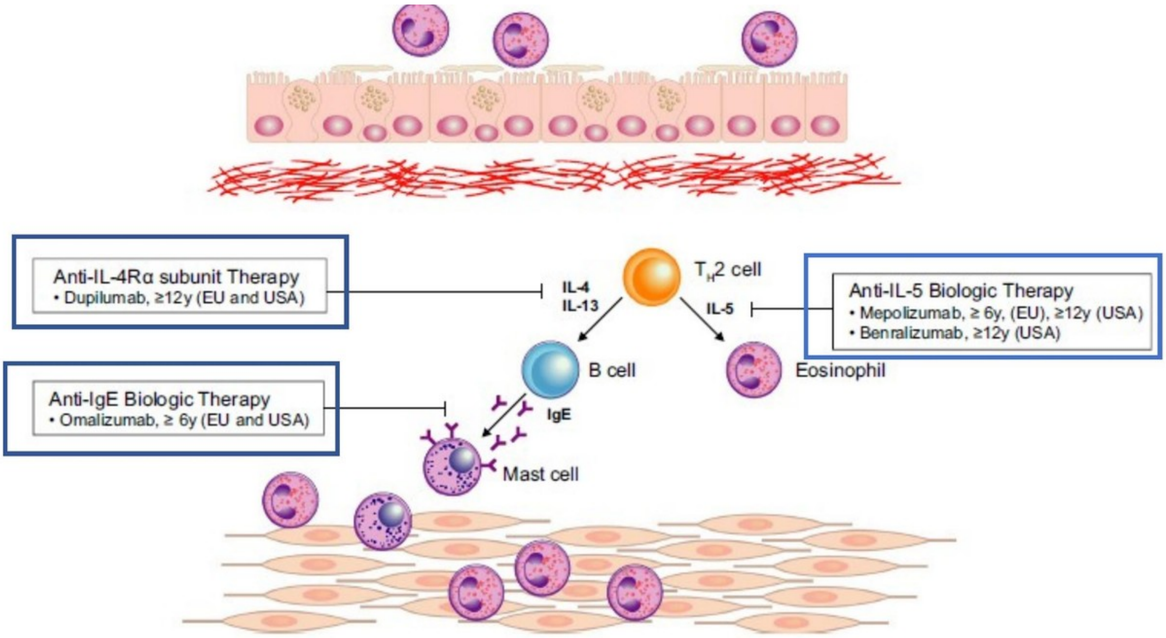

FIGURE 2 Biologics licensed for use in children with asthma. Molecular pathways and inflammatory cells that the currently licensed biologics for pediatric severe asthma target. IgE, immunoglobulin E; IL, interleukin; T_H, T Helper cell

Figure L1.24 — Biologic targets in paediatric severe asthma (Scotney 2021)